HEALTHCARE ANALYTICS | BI PROJECT

What do patient surveys from 700 hospitals tell us?

700	537K	23.8%	2.96
hospitals	completed surveys	avg response rate	avg star rating

Star schema - Python / SQL - Interactive dashboard

Star schema design

One fact table, four dimensions, 34,999 records

Dim_Location (County grain)

- Country_Id (PK)
- County / State
- City / Zip

Fact_Patient_Survey

- Country_Id (FK)
- Hospital_Id (FK)
- Measure_Id (FK)
- Survey_Id (FK)
- No_Completed_Surveys
- Response_Rate_Percent
- Linear_Mean_Value
- Answer_Percent
- Star_Rating
- + footnote columns

Dim_Measure

- Measure_Id (PK)
- Measure_Code
- Start / End date

Dim_Hospital

- Hospital_Id (PK)
- Hospital_Name
- City / Location

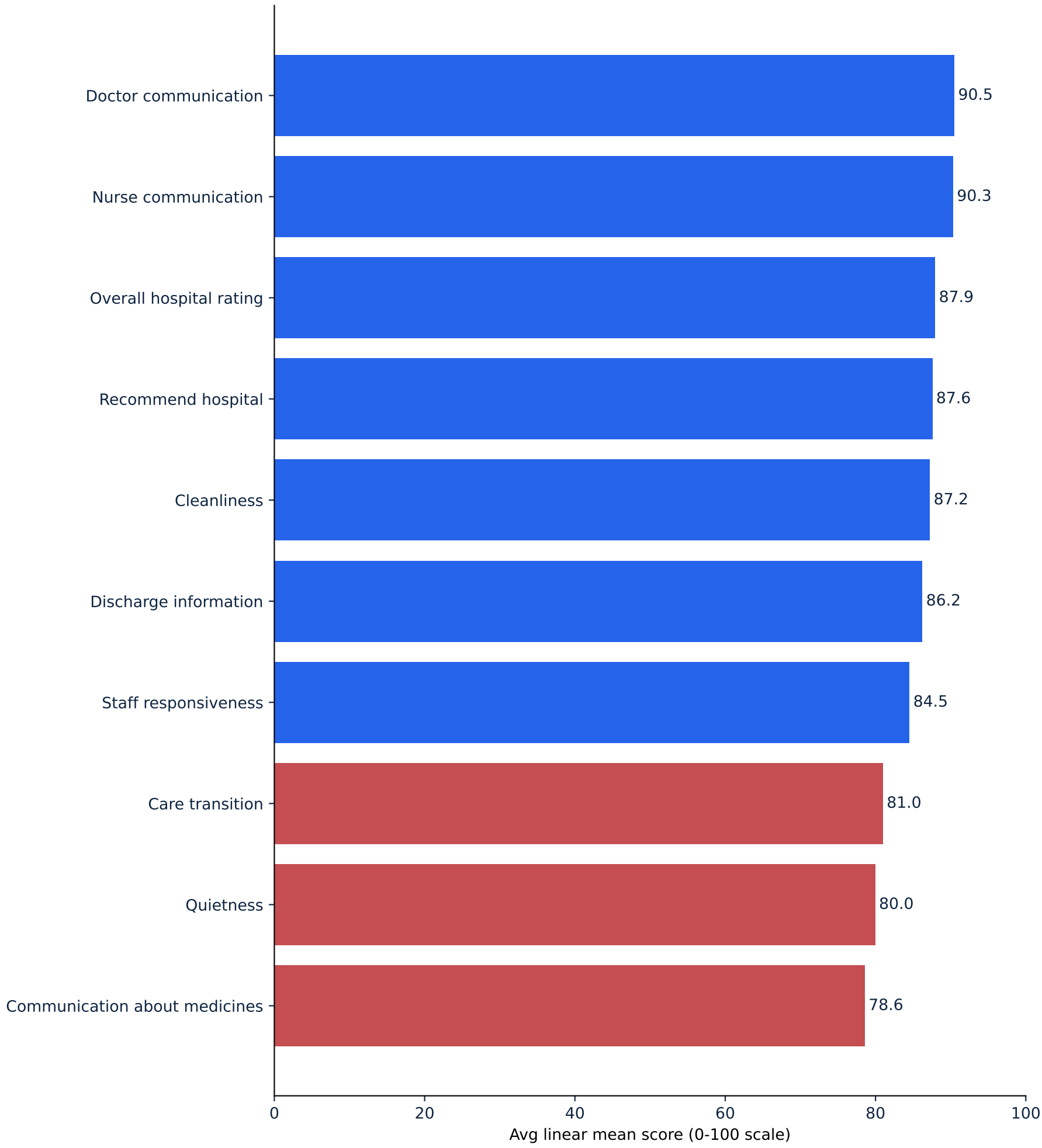
Dim_Survey

- Survey_Id (PK)
- Question
- Answer

Source: CMS HCAHPS patient survey data, 700 hospitals, 7 states, Jul 2017 - Jun 2018

Medication communication is the weakest link

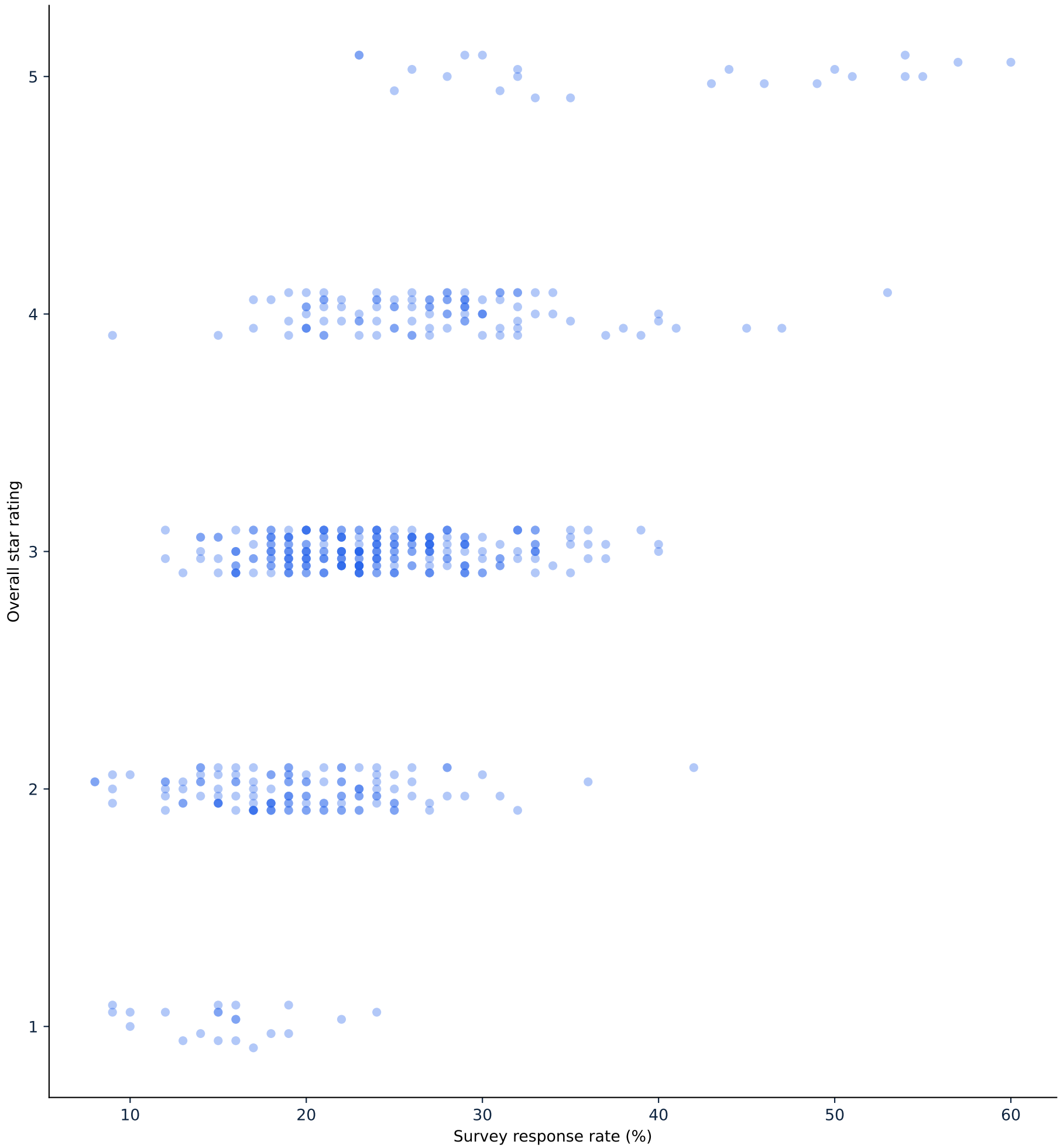
Avg HCAHPS linear mean score by category (axis starts at 0); red = lowest three



Source: CMS HCAHPS patient survey data, 700 hospitals, 7 states, Jul 2017 - Jun 2018

More responses, higher ratings

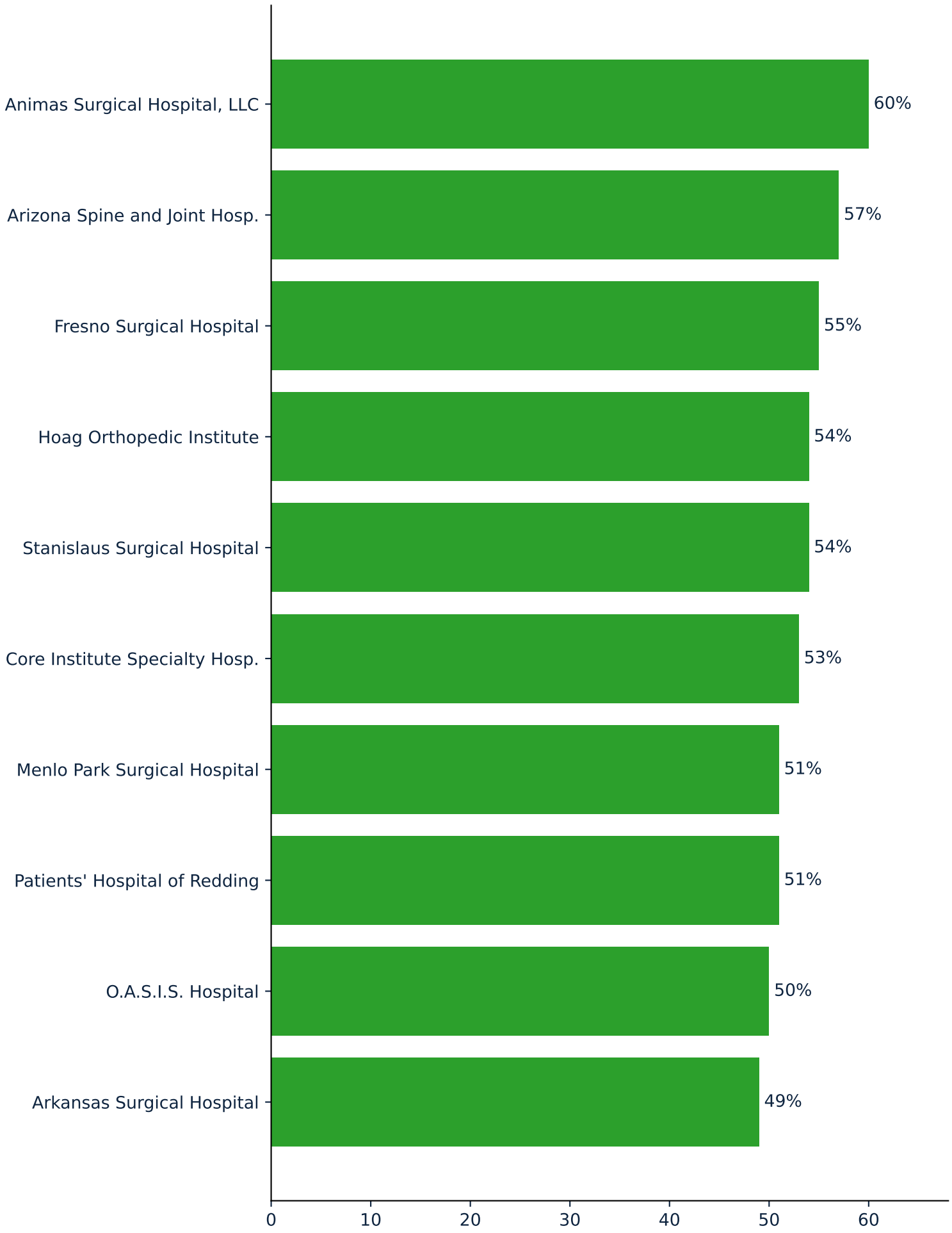
Response rate vs overall star rating per hospital (r = 0.56); correlation, not causation



Source: CMS HCAHPS patient survey data, 700 hospitals, 7 states, Jul 2017 - Jun 2018

Top 10 hospitals by survey response rate

Specialty and surgical hospitals lead on patient engagement



Source: CMS HCAHPS patient survey data, 700 hospitals, 7 states, Jul 2017 - Jun 2018

Takeaways

1. Fix medication communication first.
   It scores lowest nationally (78.6)
   vs 90+ for nurse and doctor communication.

2. Watch response rate as an early signal.
   It correlates with overall star rating
   (r = 0.56), though not proof of cause.

3. Handle thin data deliberately.
   81 small hospitals have suppressed
   results; ignoring them skews comparisons.

Source: CMS HCAHPS patient survey data, 700 hospitals, 7 states, Jul 2017 - Jun 2018